

$f: A \rightarrow B$

bool is_even(int n)

$$D_{\text{int}} = \{x : -2^{n-1} \leq x \leq 2^{n-1} \text{ where } n = \text{sizeof(int)} * 8\}.$$

$$D_{\text{bool}} = \{\text{True}, \text{False}\}.$$

$$\text{is_even} : D_{\text{int}} \rightarrow D_{\text{bool}}.$$

```
int sum(int a, int b)    (a, b) ∈ Dint × Dint
{
    return a+b;
}
```

$$D_{\text{int}} = \{x : -2^{n-1} \leq x \leq 2^{n-1} \text{ where } n = \text{sizeof(int)} * 8\}.$$

$$\text{sum} : D_{\text{int}} \times D_{\text{int}} \rightarrow D_{\text{int}}.$$

Paradigm: Procedural.

```
sum = [](int a) → std::function<int(int)>
{
    return [a](int b) → int
    {
        return a+b;
    };
};
```

$$\text{sum}(10)(20) \rightarrow 30$$

functional programming

$\text{sum} :: D_{\text{int}} \rightarrow (D_{\text{int}} \rightarrow D_{\text{int}})$ Currying



$\equiv$

$\text{sum} :: D_{\text{int}} \times D_{\text{int}} \rightarrow D_{\text{int}}$

Haskell Curry

$\text{power} :: (D_{\text{double}} \times D_{\text{int}}) \rightarrow D_{\text{double}}$

$\text{factory} :: D_{\text{int}} \rightarrow (D_{\text{double}} \rightarrow D_{\text{double}})$

$\text{std} :: \text{function} \langle \text{double} (\text{double}) \rangle$